CMSC216: Finale

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Logistics

Goals

- Threads Wrap
- Final Exam Logistics
- Review / Practice Exam
- Parting Words

Date	Event
Thu 07-May	Practice Exam
Fri 08-May	P5 Due
Sat 09-May	CourseExp Due
Sun 10-May	Optional Lab12 Due
Mon 11-May	Exit Survey Due
12-May Tue	Final Exam
6:30-8:30pm	ARM 0135

Office Hours

- Kauffman OH Mon 11-May 2-4pm
- Staff OHs end Friday, special office hours posted later

Announcements: Student Feedback Opportunities

Course Experiences Now Open

e.g. Rate your Professor

- https://www.courseexp.umd.edu/
- **If response rate reaches 80% for every section...**
- **by Sat 09-May-2026 11:59pm...**
- **I will reveal a Final Exam Question**
- No answers but public discussion welcome
  (Final Exam is on Tue 12-May-2026)

Canvas Exit Survey

- Will open on ELMS/Canvas Tomorrow
- Worth 1 Engagement Point for completion
- Due prior to Final Exam (Mon 11-May 11:59pm)

Final Exam Logistics

- Final Exam in person 6:30-8:30pm Tue 16-Dec
- In normal lecture hall: ARM 0135
- 2 hours long, Open Resource
- 6 sides of paper (3 pages front/back)
- ~3 sides like 3rd Midterm Exam
  Virtual Memory, Threads, P5 Material
- ~3 sides Comprehensive Review
  Tie together concepts that pervaded the semester like memory layout, physical parts of the computing system,
- Question format like previous exams: coding, debugging, conceptual, practical applications

What have we done?

C Programming

Lowest of the “high-level” languages, gives fairly direct control over capabilities of the machine at the expense of coding difficulty and ease of mistakes

Assembly Programming

Tied directly to what a processor can do, studied x86-64 specifically, exposes processor internals like registers, instructions, operand sizes, etc.

Computing Hardware

Basics components like CPU, Registers, Cache Memory, DRAM, Disks, how they interact

Operating System Basics

Programs exist in an environment usually managed by an OS, provides abstractions like Processes, Files, Threads, along with the ability to manipulate and coordinate these through System Calls

We studied Computing Systems, as you might expect

Further Coursework / Activities

- **CMSC411 Computer Systems Architecture**: Develops hardware/software interface in more detail, study pipelines + superscalar features in more detail, cache architecture / implementation, examine multi-core systems
- **CMSC412 Operating Systems**: Study internal design issues associated with operating systems, handling hardware, tradeoffs on different approaches to management, theoretical algorithms around resource coordination.
- **CMSC414 Computer and Network Security**: Builds on the buffer overflow attacks we studied, discusses other attacks via networks, counter measures, design aspects of secure systems
- **CMSC416 Introduction to Parallel Computing**: Hardware and Programming Models for parallelism. Builds substantially on using multiple threads/processes to cooperatively compute faster. *Kauffman will teach the Spring 2026 section.*

Off-Season Practice

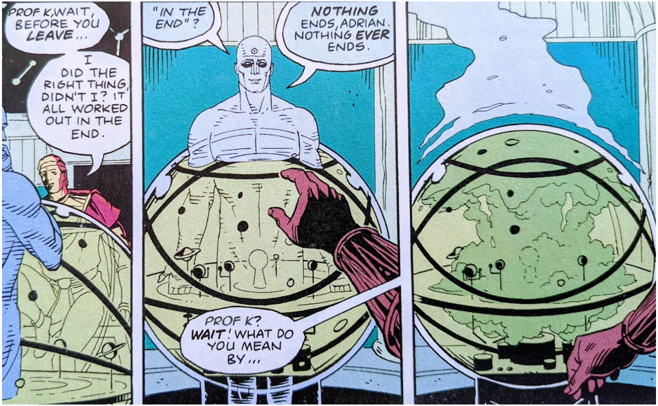
Students often ask what they could do during a break to sharpen their computing skills. Here are a few ideas.

- **Read**: The Art of Unix Programming by Eric S. Raymond
  Fantastic philosophical and pragmatic discussion of how to build systems that work especially in the Unix environment. (free online)
- **Complete**: If you didn't finish a project in this course or another, take some time to do so.
- **Extend**: If you use VS Codium, Write an Extension for it that does something interesting. This will teach you MUCH about modern software development
- **Build**: Buy an Arduino Microcontroller ($10) and get a "Blinky" routine to run; it's C code! Adafruit has tons of fun toys with accompanying tutorials.
- **Rest**: Take some time away from the screen for fun. Recharging is as important for people as for phones. Play outside. See some people in person. Breathe.

Practice Final

- Take a few minutes to look this over on your own then together
- Kauffman will answer a few questions on it and post solutions later today

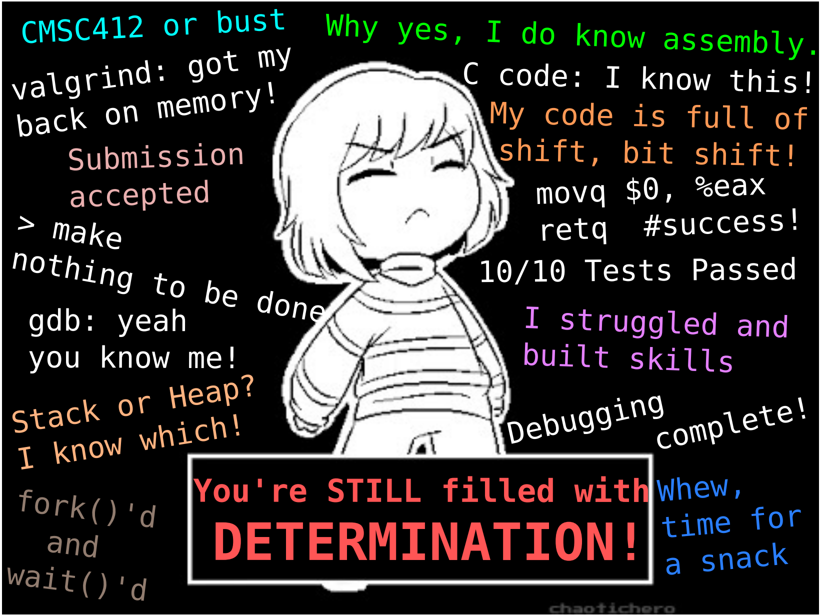
Nothing Ever Ends



- What you studied will recur in your career at some point and demonstrate whether you learned it well the first time or need another pass.
- Some of it will change in the future and make you feel old.
- Expect this and stay determined.

Conclusion

It's been a hell of a semester.
Thank you for your time,
attention, and effort.
I'm proud of all of you.
Keep up the good work.
Keep caring, about your work,
your peers, and yourself.
Take a sticker. Stay safe.
Happy Hacking.


CMSC412 or bust
Why yes, I do know assembly.
valgrind: got my back on memory!
C code: I know this!
My code is full of shift, bit shift!
Submission accepted
movq $0, %eax
retq #success!
> make
nothing to be done
10/10 Tests Passed
gdb: yeah you know me!
I struggled and built skills
Stack or Heap? I know which!
Debugging complete!
fork()'d and wait()'d
You're STILL filled with
DETERMINATION!
Whew, time for a snack
chaotichero